Functional Equations Solutions

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27 June 2026

§1 Walkthrough Problems

Problem 1.1 (USAMO 2002)**.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all pairs of real numbers x and y.

Proof. Let $P(x, y)$ denote the given assertion. Note that $P(0, 0)$ gives us $f(0) = 0$. Then,

$$\begin{aligned} P(x, -x) \implies & f(x^2 - x^2) = xf(x) + xf(-x) \\ & 0 = x(f(x) + f(-x)) \end{aligned}$$

For nonzero x, we have $f(-x) = -f(x)$, but when $x = 0$ we have $f(0) = 0$ anyway, so we have f is odd for all real numbers x.

$$P(x, 0) \implies f(x^2) = xf(x)$$

So now we can rewrite the original condition as:

$$f(x^2 - y^2) = f(x^2) - f(y^2)$$

We let $a = x^2$, and $b = y^2$ such that, for nonnegative a, b we have:

$$f(a - b) = f(a) - f(b)$$

Again, we can set $a = x_0 + y_0$ and $b = y_0$ to get:

$$f(x_0 + y_0 - y_0) = f(x_0 + y_0) - f(y_0) \implies f(x_0 + y_0) = f(x_0) + f(y_0)$$

for nonnegative x_0, y_0. However, since f is odd, we get that f is additive on the negatives too.

Finally, we let $x = a + b$ in one of the previous equations $f(x^2) = xf(x)$ to get:

$$f(a^2 + 2ab + b^2) = (a + b)(f(a) + f(b))$$

$$f(2ab) = af(b) + bf(a)$$

$$a = 1 \implies f(2b) = f(b) + f(1)b = 2f(b)$$

Which gives $f(x) = cx$ for all x, which upon checking works. $\square$

Problem 1.2 (IMO 2017)**.** Solve over $\mathbb{R}$ the functional equation

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

Proof. We claim that $f(x) = 0$, $f(x) = 1 - x$, and $f(x) = x - 1$ are the only solutions. It's easy to check that these solutions indeed work.

Let $P(x, y)$ denote the given assertion. First, note:

$$P(0,0) \implies f(f(0)^2 = 0) \implies \text{ there exists a } z \text{ such that } f(z) = 0.$$

Also, $P(x, 0)$ gives that $f(0) = 0 \implies f \equiv 0$. So from now assume that $f(0) \neq 0$. We take, for $z \neq 1$,

$$P\left(z, \frac{z}{z-1}\right) \implies f\left(f(z)f\left(\frac{z}{z-1}\right)\right) = 0$$

giving $f(0) = 0$, which is a contradiction, giving $z = f(0)^2 = 1 \implies f(0) = \pm 1$. One can now take $P(1, x)$ to give $f(x+1) = f(x) - 1$, then $P(x, 0)$ to get $f(f(x)) = 1 - f(x)$. Applying f on this equation and equating some terms gives us:

$$f(f(f(x))) = f(1 - f(x)) = 1 - f(f(x)) = f(x)$$

Before we move on, I'd like to prove something that would help us in our final claim.

Claim 1.3 — For sufficiently large N_1 and N_2, we can find x, y satisfying $x + y = N_1 + 1$ and $xy = N_2$.

Proof. First we observe that we can think of x, y as roots of some quadratic equation. Of course, comparing it with Vieta's gives us that x and y are possible roots of the equation

$$x_0^2 - (N_1 + 1)x_0 + N_2 = 0$$

Obviously distinct solutions x, y to this equation exist if and only if the discriminant, Δ is greater than zero. So we have:

$$\begin{aligned} \Delta &= (N_1 + 1)^2 - 4N_2 \\ &= N_1^2 + 2N_1 + 1 - 4N_2 > 0 \end{aligned}$$

So we can just pick sufficiently large N_1, N_2 that satisfies the condition(or maybe something like $N_1^2 \geq 4N_2$) so we have solutions for x, y. $\square$

Now, we prove the final claim.

Claim 1.4 — f is injective.

Proof. Assume for some sufficiently large a, b, we have $f(a) = f(b)$ and also, $x + y = a + 1$ and $xy = b$(from the above claim, we can do this). Then, the original assertion gives us

$$\begin{aligned} f(f(x)f(y)) &= f(b) - f(a+1) \\ &= f(b) - (f(a) - 1) \\ &= 1 \end{aligned}$$

Shifting f we get that $f(f(x)f(y) + 1) = 0 \implies f(x)f(y) + 1 = 1$ so $f(x)f(y) = 0$, giving that $1 \in \{x, y\}$. However, we are done as if $x = 1$ we get $y + 1 = a + 1 \implies y = a$ and $xy = b \implies b = y = a$. This follows similarly for $y = 1$, and so f is injective. $\square$

From the injectivity, and our earlier calculations we get $f(1 - f(x)) = f(x) \implies f(x) = 1 - x$ and since $-f$ must be a solution too, $f(x) = x - 1$ is another solution. Therefore, our solutions are:

$$\boxed{f(x) = 0, 1 - x, x - 1}$$

$\square$